

SECTION 5.4: COMPARISON TESTS EXTRA PRACTICE

A. $\sum_{n=1}^{\infty} \frac{1}{n2^n}$ Compare to $\sum \left(\frac{1}{2}\right)^n$, a convergent geometric series ($|r| = \frac{1}{2} < 1$)

Since $n2^n > 2^n$, $\frac{1}{n2^n} < \frac{1}{2^n}$. So by the Comparison Test,

$\sum \frac{1}{n2^n}$ Converges. (L.C.T. works just as well...)

B. $\sum_{n=1}^{\infty} 2^n$ Diverges by the Divergence Test.

$$\lim_{n \rightarrow \infty} 2^n = \infty \neq 0.$$

C. $\sum_{n=1}^{\infty} \frac{n-4}{n^3+2n}$. Limit Comparison Test using $\sum_{n=1}^{\infty} \frac{1}{n^2}$, a convergent p-series. ($p=2 > 1$.)

$$\lim_{n \rightarrow \infty} \frac{n-4}{n^3+2n} \cdot \frac{n^2}{1} = \lim_{n \rightarrow \infty} \frac{n^3-4n^2}{n^3+2n} = 1. \text{ Thus, } \sum_{n=1}^{\infty} \frac{n-4}{n^3+2n} \text{ converges.}$$

D. $\sum_{n=2}^{\infty} \frac{1+\cos(n)}{e^n}$ Comparison to $\sum_{n=1}^{\infty} 2\left(\frac{1}{e}\right)^n$, a convergent geometric series (i.e. $|r| = \frac{1}{e} < 1$).

Since $-1 \leq \cos(n) \leq 1$, $0 \leq 1+\cos(n) \leq 2$. So

$\frac{1+\cos(n)}{e^n} \leq \frac{2}{e^n}$. Thus, by C.T., $\sum \frac{1+\cos(n)}{e^n}$ converges

L.C.T. problematic :)

E. $\sum_{n=2}^{\infty} \frac{1}{n!}$ Comparison Test using $\sum \frac{1}{n^2}$, a convergent p-series

check

n	$n=1$	$n=2$	$n=3$	$n=4$	$n=5$
$n!$	1	2	6	24	120
n^2	1	4	9	16	25

$$n! > n^2 \text{ if } n \geq 4$$

$$\text{So } \frac{1}{n!} < \frac{1}{n^2} \text{ if } n \geq 4$$

Since $\sum \frac{1}{n^2}$ converges and $\frac{1}{n!} < \frac{1}{n^2}$ for $n \geq 4$,
the series $\sum \frac{1}{n!}$ converges.

F. $\sum_{n=1}^{\infty} \frac{n}{n^2+1}$ Limit Comparison Test to $\sum_{n=1}^{\infty} \frac{1}{n}$, a divergent Series

$$\lim_{n \rightarrow \infty} \frac{n}{n^2+1} \cdot \frac{n}{1} = \lim_{n \rightarrow \infty} \frac{n^2}{n^2+1} = 1.$$

Thus, by the L.C.T., $\sum_{n=1}^{\infty} \frac{n}{n^2+1}$ converges

G. $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^3}$ Try the Integral test.

$$\int_2^{\infty} \frac{(\ln x)^{-3}}{x} dx = \lim_{t \rightarrow \infty} \int_2^t \frac{(\ln x)^{-3}}{x} dx = \lim_{t \rightarrow \infty} \left[-\frac{1}{2} (\ln x)^{-2} \right]_2^t$$

$$= \lim_{t \rightarrow \infty} \left[\underbrace{-\frac{1}{2} (\ln(t))^2}_{\downarrow 0} + \frac{1}{2} (\ln(2))^2 \right] = \frac{1}{2(\ln 2)^2}.$$

So the series converges.

H.
$$\sum_{n=1}^{\infty} \frac{n}{2^n}$$

I.
$$\sum_{n=3}^{\infty} \frac{n^2}{\sqrt{n^3-1}}$$

J.
$$\sum_{n=1}^{\infty} \frac{n^3}{(n^4-3)^2}$$

$$\mathbf{K.} \quad \sum_{n=1}^{\infty} (-1)^n 3^{-n/3}$$

$$\mathbf{L.} \quad \sum_{n=2}^{\infty} \frac{1}{n^2 - 1}$$